Sharper Mesh-Only Convergence Rates

Recovered ChatGPT argument · 17 August 2026

This note preserves the argument from the conversation *Orthogonal Tiling Convergence Proof*, in response to “do you think you can derive a sharper result?”. It was recovered and typeset on 11 October 2026. Mathematical content is transcribed below; display layout and section styling have been adapted.

Read with *Mesh-Only Convergence for Orthogonal Tilings*. Proposition numbers in the transcript refer to the original draft. The later rewritten manuscript retains the weaker exponent 4/(3d + 8); the sharper argument was not incorporated into that PDF.

Scope: the finite Dirichlet setup of the companion note, in dimension at least two. The domain U is compactly contained in the ambient tiled domain; h is harmonic and twice continuously differentiable on a fixed neighborhood K of the closure of U. The discrete extension has matching boundary values, and the mesh is the largest cell diameter. The rate concerns uniform harmonic-extension error, not a quantitative rate for convergence of parametrized walks. The inputs are Proposition 2.3, Proposition 3.1, Lemma 4.2, and Lemma 5.1 of the rewritten note; this transcript is not a standalone proof of those inputs.

Original argument

Yes. The previous rate $\varepsilon^{4/(3d+8)}$ is not the best consequence of the argument. A sharper use of Proposition 2.3 gives

$$\|h_n^D - h\|_{L^\infty(V_n[U])} \le C\|h\|_{C^2(K)}\varepsilon_n^{4/(d+4)}.$$

The key improvement is to estimate the expected exit time of a small-volume set rather than choosing a deterministic time.

1. Expected exit time of a set

Let S be an interior finite vertex set and write

$$R := \Big(\sum_{v\in S} \mathrm{Vol}_d(P_v)\Big)^{1/d}.$$

Proposition 2.3 implies that the first Dirichlet eigenvalue on S satisfies

$$\lambda_1(S) \ge \frac{c_d}{R^2+\varepsilon R}. \tag{1}$$

Set

$$\rho := R^2 + \varepsilon R.$$

Proposition 3.1 gives

$$p^S_{2\rho}(x,x) \le C_d(\rho^{-d/2} + \varepsilon^d \rho^{-d}).$$

Since $\mathrm{Vol}(S) = R^d$,

$$\begin{aligned}\mathrm{Vol}(S) p_{2\rho}^S(x,x) &\le C_d[(\tfrac{R^2}{\rho})^{d/2} + (\tfrac{\varepsilon R}{\rho})^d] \\ &\le C_d.\end{aligned} \tag{2}$$

For $t \ge \rho$, the L^2 spectral contraction and (2) give

$$\mathbb{P}_x(\tau_S > t) \le C_d \exp(-\tfrac{c_d(t-\rho)}{\rho}).$$

Integrating the survival probability,

$$\sup_{x \in S} \mathbb{E}_x[\tau_S] \le C_d(R^2 + \varepsilon R). \tag{3}$$

This is the discrete analogue of the fact that Brownian motion exits a set of volume R^d in expected time at most CR^2.

2. Apply this to a superlevel set

Fix $\eta > 0$ and set

$$S_n := \{v : h_n^D(v) - h(v) > \eta/2\}.$$

The L^2 estimate gives

$$\sum_{v \in S_n} \mathrm{Vol}_d(P_v) \le C \|h\|_{C^2(K)}^2 \frac{\varepsilon_n^2}{\eta^2}.$$

After normalizing $\|h\|_{C^2(K)} \le 1$, this means

$$R_n \le C(\tfrac{\varepsilon_n}{\eta})^{2/d}. \tag{4}$$

Fix a small radius $r > 0$, independent of n, and stop the walk at

$$\sigma = \tau_{S_n} \wedge \tau_{B_r(z)}.$$

The coordinate functions are discrete harmonic, so

$$\mathbb{E}_z[X_\sigma - z] = 0. \tag{5}$$

The exact clock identity and (3) give

$$\mathbb{E}_z \|X_\sigma - z\|^2 \le C(R_n^2 + \varepsilon_n R_n + \varepsilon_n). \tag{6}$$

Taylor's theorem, together with (5), gives

$$\mathbb{E}_z[h(X_\sigma) - h(z)] \le C \|\nabla^2 h\|_\infty \mathbb{E}_z \|X_\sigma - z\|^2. \tag{7}$$

Furthermore,

$$\mathbb{P}_z(\tau_{B_r(z)} < \tau_{S_n}) \le \tfrac{1}{r^2}\mathbb{E}_z\|X_\sigma - z\|^2. \tag{8}$$

Optional stopping for h_n^D, using (7)–(8), therefore yields

$$\sup_{v \in V_n[U]} (h_n^D(v) - h(v)) \le C(\eta + R_n^2 + \varepsilon_n R_n + \varepsilon_n).$$

Applying the same argument to $(-h_n^D,\ -h)$,

$$\|h_n^D - h\|_\infty \le C[\eta + (\tfrac{\varepsilon_n}{\eta})^{4/d} + \varepsilon_n(\tfrac{\varepsilon_n}{\eta})^{2/d} + \varepsilon_n]. \tag{9}$$

3. Optimize

Balance the first two terms:

$$\eta = (\tfrac{\varepsilon_n}{\eta})^{4/d}.$$

This gives

$$\eta = \varepsilon_n^{4/(d+4)}.$$

The remaining terms in (9) are smaller. Hence

$$\|h_n^D - h\|_\infty \le C_{U,K,d}\|h\|_{C^2(K)}\varepsilon_n^{4/(d+4)}.$$

For example,

$$d = 2:\ \varepsilon_n^{2/3}, \qquad d = 3:\ \varepsilon_n^{4/7}, \qquad d = 4:\ \varepsilon_n^{1/2}.$$

This is likely the natural rate obtainable from the combination of an L^2-error of order ε and diffusive exit-time control.